

Aliah University

Odd (Autumn)-Semester Examination – 2023 -24

(For 5th Semester B.Tech(CSE) Programme)

Paper Name: Operating System

Full Marks: 80

Paper Code: CSEUGPC10

Time: 3 Hrs

Group – A

(5X2=10)

1. Answer all questions –

- a) What is Kernel?
- b) What is a process control block (PCB)?
- c) What is race condition?
- d) State necessary conditions for a deadlock situation to arise.
- e) What is internal and external fragmentation?

Group – B

(Answer any 6 questions)

(6X5=30)

2. What is a process? Draw and explain the process state diagram. 1 + 4
3. Explain the difference between long term, short term and medium term schedulers. 5
4. What is the critical section? What requirement should be satisfied for a solution to the critical section problem? 1 + 4
5. Discuss the Deadlock detection algorithm with an example. 5
6. What is paging? Discuss basic paging techniques in detail. 1 + 4
7. Memory partitions of 100kb, 500kb, 200kb, 300kb, 600 kb are available. What will be the performance of best, worst and first fit algorithm, assuming the processes are coming in the following order - 212,417,112,426 ? 5
8. What is demand paging? Explain it with the address translation mechanism used. What are its specific advantages? 1 + 3 + 1
9. Explain various file allocation methods. 5

Group – C

(Answer any 4 questions)

(10X4=40)

10. Explain in detail various types of operating systems. 10
11. Consider the following set of process with the length of CPU burst time given in milliseconds: 10

Process	Arrival Time	Burst Time	Priority
P1	3	3	1
P2	4	4	2
P3	1	5	4
P4	5	3	2
P5	2	2	3

Draw the four Gantt charts illustrating the execution of these processes using FCFS, SJF, Shortest-remaining-time-first, Preemptive priority, Round Robin (quantum = 2) scheduling and calculate waiting and turnaround time for each process for each algorithm.

12. What are semaphores? Explain two primitive semaphore operations. State dining philosopher's problem and give a solution using semaphores. 1 + 2 + 7

13. Consider the following snapshot of a system

Process	Allocation A B C D	Max A B C D	Available A B C D
P1	2 0 0 1	4 2 1 2	3 3 2 1
P2	3 1 2 1	5 2 5 2	
P3	2 1 0 3	2 3 1 6	
P4	1 3 1 2	1 4 2 4	
P5	1 4 3 2	3 6 6 5	

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Illustrate that the system is in a safe state by demonstrating an order in which the processes may complete. If a request from process P2 arrives for (1, 1, 0, 0), can the request be granted immediately? 5 + 5

14. What is the page fault? For the page reference string 7,0,1,2,0,3,0,4,2,3,0,3,0,3,2,1,2,0,1,7,0,1, find out the number of page faults using FCFS, Optimal and LRU page replacement policies. 15, 12, 9 1 + 9

15. Illustrate the disk scheduling algorithms FCFS, SSTF, SCAN, C-SCAN, and C - LOOK with a request queue (0-199) and the current head pointer is at 100. 10
98, 183, 37, 122, 14, 124, 65, 67

$$\begin{array}{r}
 183 \\
 - 98 \\
 \hline
 105 \\
 183 \\
 \hline
 288 \\
 - 37 \\
 \hline
 251 \\
 244 \\
 \hline
 495 \\
 - 51 \\
 \hline
 444 \\
 + 148 \\
 \hline
 592 \\
 - 79 \\
 \hline
 513
 \end{array}$$

513
+

$$\begin{array}{r}
 124 \\
 98 \\
 \hline
 26 \\
 - 2 \\
 \hline
 124 \\
 183 \\
 \hline
 307 \\
 2 \\
 \hline
 \end{array}$$